

Epistatic Circuits: Exhaustive Interaction Decomposition of Transformer Circuits

Anonymous authors

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Abstract

Transformer circuit discovery methods rank components by individual importance scores and treat the resulting circuits as additive. Yet attention mechanisms are structurally incapable of implementing additive models: the softmax normalization forces every head’s output to depend on every other head in the same layer. We provide the first exact, non-human-curated ground truth for pairwise interaction structure in a real trained circuit. The ground truth comes from exhaustive Walsh–Hadamard coalition sweeps over 15- and 20-head subcircuits of GPT-2 small. We then use this ground truth to benchmark three discovery methods. Weight-space geometry (OV and QK composition scores) carries no cross-validated predictive power for pairwise interaction (CV $R^2 < 0$ across all conditions). Edge Attribution Patching (EAP) agrees moderately with the Walsh interaction spectrum (Spearman $\rho = 0.58$). Path patching, which isolates direct causal edges, measures a nearly orthogonal quantity (Spearman $\rho = 0.21$, signed Pearson $r = 0.07$). Compressed sensing makes the ground truth practical to compute: LASSO recovers the full pairwise spectrum from 140 random coalitions ($r > 0.95$, $7,490\times$ compression). The recovered interaction structure carries mechanism — a complementation test on the interaction matrix rediscovers that negative name movers invert sign when their upstream S-inhibition heads are removed, a dependency the original IOI analysis flagged as unexplained. Each discovery method imposes a different selective pressure on which interactions survive, and the choice of method shapes the circuit one finds.

1 Introduction

Circuit discovery in transformers identifies a small subnetwork that explains a model’s behavior on a target task. The dominant methods — activation patching (Wang et al., 2023), Edge Attribution Patching (EAP) (Syed et al., 2023), and Automatic Circuit Discovery (ACDC) (Conmy et al., 2023) — rank components by individual importance and assemble circuits by thresholding those scores. The resulting circuits are additive by construction: each component contributes independently, and the circuit’s total effect is the sum of its parts.

Attention mechanisms violate this assumption at the architecture level. Leemann et al. (2024) prove that transformers with attention mechanisms structurally cannot represent additive models over sequences of more than one token: the softmax normalization forces every head’s output to depend on every other head in the same layer. The theorem applies to both single-layer and multi-layer architectures, and the result is tight — the non-additivity is a property of the function class, not a failure of training. Any circuit discovery method that reports only marginal importance scores discards this interaction structure by design.

The gap between additive discovery and non-additive computation raises a concrete question: how much interaction structure do different discovery methods retain, and does the discarded structure carry mechanism? Answering the question requires exact ground truth for the interaction between every pair of components in a circuit — data that, until this work, has not been available for any real trained model.

We provide this ground truth by computing exhaustive Walsh–Hadamard coalition sweeps over subcircuits of GPT-2 small on the indirect object identification (IOI) task. The Walsh–Hadamard transform decomposes the 2^n coalition function values into a spectrum of interaction coefficients, where each order-2 coefficient w_{ij}

measures how much the joint effect of heads i and j deviates from the sum of their individual effects. The decomposition is exact, reference-free, and requires no regularization or held-out validation — properties inherited from the transform’s orthogonality.

Three results emerge from benchmarking discovery methods against this ground truth:

1. **Weight geometry predicts nothing.** OV composition, QK composition, and layer distance — the standard weight-space features from Elhage et al. (2021) — carry no cross-validated predictive power for pairwise interaction coefficients (Section 3.3). Interaction is a property of the forward pass, not the weight matrices.
2. **Different methods recover different structure.** EAP edge scores agree moderately with the Walsh interaction spectrum (Spearman $\rho = 0.58$), capturing the linear component of pairwise interaction. Path patching, which isolates direct causal edges through the residual stream, measures a nearly orthogonal quantity (signed Pearson $r = 0.07$). The two methods impose different selective pressures on which interactions survive (Section 3.4).
3. **The interaction structure carries mechanism.** Applying a complementation test from genetics to the Walsh interaction matrix rediscovers that negative name movers invert their contribution sign when their upstream S-inhibition heads are removed — a dependency that Wang et al. (2023) observed as unexplained (Section 4).

Compressed sensing makes the ground truth practical to compute beyond small circuits: LASSO recovers the full pairwise interaction spectrum of a 20-head circuit from 140 random coalitions ($r > 0.95$), a $7,490 \times$ compression over exhaustive enumeration (Section 3.1). For 30-head circuits, the cost is roughly 500 forward passes — a ten-minute GPU job rather than an impossible 2^{30} sweep.

The selective-pressure framing applies beyond the methods tested here. Any circuit discovery method that reports component-level scores can be evaluated against the Walsh ground truth by asking how much of the pairwise interaction spectrum it preserves. The interaction spectrum itself is a fixed property of the circuit and task, independent of how it was discovered.

2 Background

2.1 Walsh–Hadamard interaction decomposition

Given a circuit of n components (attention heads), the coalition function $v : \{0, 1\}^n \rightarrow \mathbb{R}$ maps each subset of active heads to the model’s performance metric (logit difference) when those heads operate normally and the rest are ablated. The 2^n function values admit a unique Walsh–Hadamard decomposition

$$v(x) = \sum_{S \subseteq [n]} w_S \chi_S(x), \quad \chi_S(x) = (-1)^{\sum_{i \in S} x_i}, \quad (1)$$

where the sum ranges over all 2^n subsets S of $[n] = \{1, \dots, n\}$ and χ_S is the corresponding Walsh basis function (Poelwijk et al., 2016). The coefficient w_S is the unique order- $|S|$ interaction among the heads in S :

$$w_S = \frac{1}{2^n} \sum_{x \in \{0, 1\}^n} v(x) \chi_S(x). \quad (2)$$

Order-1 coefficients $w_{\{i\}}$ measure each head’s marginal importance averaged over all 2^{n-1} contexts. Order-2 coefficients $w_{\{i, j\}}$ measure how much the joint effect of heads i and j deviates from the sum of their individual effects, again averaged over all contexts. The decomposition is reference-free: unlike pairwise epistasis measured against a single wildtype, each Walsh coefficient averages the relevant interaction over all possible backgrounds, eliminating dependence on an arbitrary reference state (Poelwijk et al., 2019).

The transform satisfies Parseval’s identity: $\sum_S w_S^2 = 2^{-n} \sum_x v(x)^2$. The fraction of total variance (“energy”) carried by each order quantifies how much of the circuit’s behavior requires combinations of that many

heads. A circuit whose energy concentrates at order 1 is well approximated by an additive model; one with substantial order-2 energy has pairwise interactions that an additive summary discards.

2.2 Ablation primitives

The coalition function depends on how inactive heads are ablated. Three primitives are standard:

- **Zero ablation:** replace the head’s output (`hook_z`) with zero. Removes the head’s contribution to the residual stream entirely.
- **Mean ablation:** replace `hook_z` with its mean across prompts. Substitutes the head’s average behavior, preserving the residual stream’s scale.
- **Resample ablation:** replace `hook_z` with its value on a different prompt from the same distribution. Preserves distributional statistics while destroying task-specific information.

Different primitives yield quantitatively different Walsh spectra. We report results under all three where exhaustive sweeps are available (15-head circuits) and under mean ablation for the 20-head circuit.

2.3 Circuit discovery methods

Activation patching. The marginal importance of each head is measured by ablating it in isolation and recording the change in the target metric. Activation patching by mean ablation on `hook_z` is the standard protocol (Wang et al., 2023). The resulting scores are order-1 quantities: they measure each head independently, discarding all interaction structure.

Edge Attribution Patching (EAP). Syed et al. (2023) approximate the effect of ablating each edge (sender–receiver pair) by backpropagating through the clean forward pass and multiplying gradients by activation differences. The resulting edge scores are linearized interaction estimates: they capture how much one head’s output affects another’s computation, limited to first order. EAP runs in a single forward–backward pass and scales to large circuits.

Path patching. Path patching isolates the direct causal edge from sender S to receiver R by corrupting S , freezing all intermediate components to their clean values, and letting R compute naturally from the modified residual stream (Wang et al., 2023). The change in the target metric relative to the clean run is the direct effect of S on the output routed specifically through R . Path patching distinguishes direct edges from mediated interactions but requires one forward pass per edge.

2.4 MIB faithfulness metrics

The Mechanistic Interpretability Benchmark (MIB) (Mueller et al., 2025) evaluates circuit discovery methods by how faithfully a discovered circuit reproduces the full model’s behavior. Given a metric m (such as logit difference), the faithfulness of a circuit C relative to the full model N is

$$f(C, N; m) = \frac{m(C) - m(\emptyset)}{m(N) - m(\emptyset)}, \quad (3)$$

where $m(\emptyset)$ is the metric with all components ablated and $m(N)$ is the metric with no ablation. A faithfulness of 1 means the circuit fully reproduces the model’s behavior; 0 means the circuit performs no better than complete ablation.

Two summary statistics aggregate the faithfulness curve across circuit sizes:

- **CPR** (Circuit Performance Recovered): the area under the faithfulness curve as a function of circuit size, normalized to $[0, 1]$. Higher is better.
- **CMD** (Circuit–Model Distance): the area between the faithfulness curve and the ideal curve $f = 1$, normalized to $[0, 1]$. Lower is better. $\text{CMD} = 1 - \text{CPR}$ when faithfulness is monotone.

3 Results

3.1 Sparse recovery scales coalition sweeps beyond 15 heads

A coalition sweep over n heads requires 2^n forward passes: 32 768 for 15 heads, feasible but slow; $\sim 10^6$ for 20 heads, borderline; $\sim 10^9$ for 30, impossible. The exponential cost confines Walsh interaction analysis to small circuits. Compressed sensing (Candès & Tao, 2005) offers an alternative: sample $M \ll 2^n$ random coalitions, build the corresponding rows of the Walsh basis matrix $\Phi_{mc} = (-1)^{\text{popcount}(c \& s)}$, and solve a sparse regression for the $k = n + \binom{n}{2}$ order-1 and order-2 coefficients. If the interaction spectrum is compressible, recovery succeeds at M far below 2^n .

Theoretical brackets. Two sample-complexity regimes bound the required M . The generic RIP bound for Gaussian or Bernoulli measurement matrices gives $M = O(k \log(N/k))$ (Candès & Tao, 2005); for $k = 210$ and $N = 2^{20}$, this is $M \approx 1400$. Our measurement matrix is a subsampled Walsh–Hadamard matrix, where the tighter structured bound $M = O(k \log k \log N)$ applies (Bourgain, 2014; Rudelson & Vershynin, 2008), with a matching lower bound $\Omega(k \log k \log(N/k))$ from Blasiok et al. (2019). For the same parameters this gives $M \approx 13\,000$. Recovery should become reliable somewhere between these bounds.

Methods. Three recovery algorithms were compared. LASSO with cross-validated regularization (LassoCV, 5-fold) selects both which coefficients are nonzero and their magnitudes. Orthogonal matching pursuit with cross-validated sparsity (OrthogonalMatchingPursuitCV, 5-fold) makes the same selection greedily. Neither method receives oracle knowledge of k . Monte Carlo direct estimation, $\hat{w}_S = M^{-1} \sum_c f(c) \chi_S(c)$, serves as an unbiased but high-variance baseline.

Phase 1: validation on existing exact sweeps. Fourteen completed exact sweeps (15-head IOI and RTI circuits under three ablation primitives, $k = 120$, $N = 32\,768$) provided ground truth. At each sample size M , ten random subsets of the full 2^n coalition values were drawn, and sparse recovery was compared to the exact Walsh spectrum. LASSO achieved $r > 0.95$ at $M = 100$ ($M/k = 0.83$) and $r > 0.99$ at $M = 500$ ($M/k = 4.2$) across all fourteen circuits, with OMP trailing by a small margin. Monte Carlo required $M > 2000$ for $r > 0.85$.

Phase 2: a 20-head circuit where exact sweeps are infeasible. The top 20 attention heads for IOI in GPT-2 small were identified by activation patching (mean ablation of each of 144 heads on 200 IOI prompts, ranked by change in logit difference). The resulting circuit has $k = 20 + \binom{20}{2} = 210$ target coefficients and $2^{20} \approx 10^6$ possible coalitions. Two thousand random coalitions were evaluated (each requiring one forward pass with a random subset of the 20 heads mean-ablated), and pairwise Walsh coefficients were recovered via LASSO and OMP.

M	M/k	Compression	LASSO r	OMP r	MC r	Top-10 recall
100	0.48	10 486×	0.91 ± 0.02	0.85 ± 0.04	0.24 ± 0.07	83%
140	0.67	7 490×	0.96 ± 0.01	0.94 ± 0.02	—	91%
200	0.95	5 243×	0.98 ± 0.00	0.98 ± 0.01	0.33 ± 0.04	93%
500	2.38	2 097×	1.00 ± 0.00	1.00 ± 0.00	0.50 ± 0.04	97%
2000	9.52	524×	1.00 ± 0.00	1.00 ± 0.00	0.77 ± 0.00	100%

Recovery reaches $r > 0.95$ at $M = 140$ ($M/k = 0.67$), meaning fewer random coalitions are needed than there are coefficients to recover. The effective sparsity of the interaction spectrum is far below the nominal $k = 210$: LASSO selects roughly 50–80 nonzero coefficients at the threshold M , and the remaining coefficients are small enough that regularization correctly zeros them. Split-half validation — two independent LASSO recoveries from disjoint 1000-coalition subsets — yields $r = 0.997$ between the two estimates.

Figure 1(a) traces the full recovery curve from $M = 20$ to $M = 2000$. LASSO dominates OMP at small M and the two converge above $M/k \approx 1$. Monte Carlo remains below $r = 0.8$ even at $M = 2000$, confirming that sparse regression is essential — random sampling alone does not exploit the compressibility. Figure 1(b)

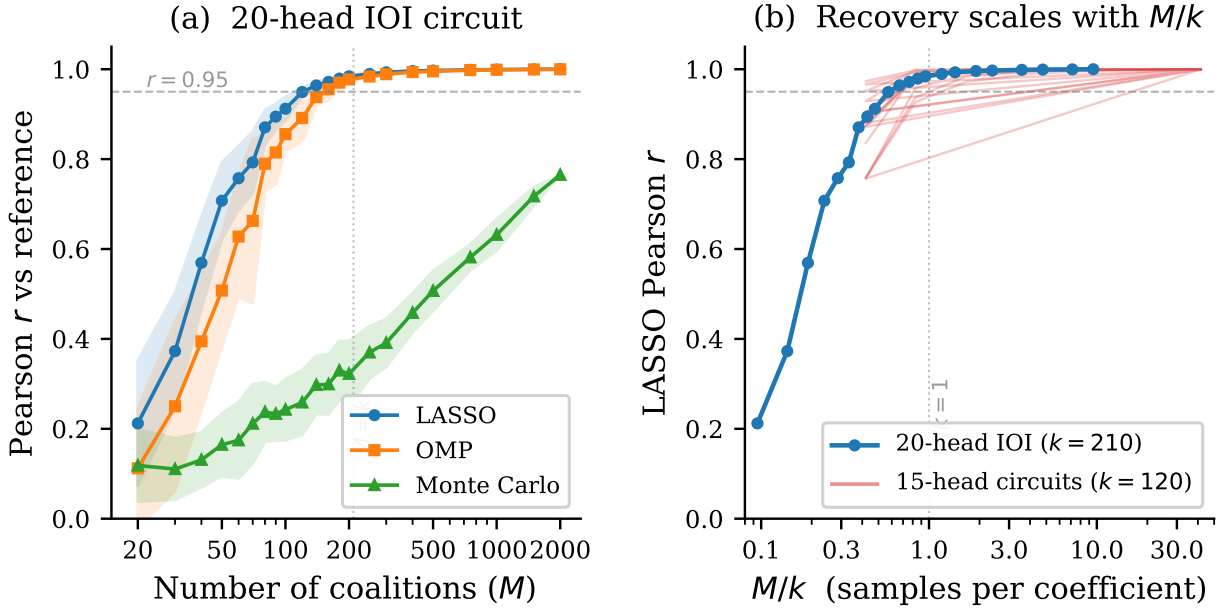


Figure 1: Sparse Walsh recovery of pairwise interaction coefficients. **(a)** Recovery quality (Pearson r vs. $M=2000$ reference) for the 20-head IOI circuit in GPT-2 small. LASSO crosses $r = 0.95$ at $M = 140$ ($7490\times$ compression over exact enumeration). Shaded bands show ± 1 s.d. across 20 random trials. **(b)** LASSO recovery across circuits of different sizes, plotted against M/k (samples per target coefficient). Fourteen 15-head circuits from Phase 1 (light lines) and the 20-head circuit (dark line) follow the same curve, confirming that the sample complexity scales with the number of coefficients, not with the number of coalitions.

overlays the 20-head and 15-head recovery curves as a function of M/k . The curves collapse: recovery quality depends on samples per coefficient, not on circuit size.

Walsh interaction vs. EAP edge attribution. The recovered Walsh coefficients measure actual coalition interaction: how much the joint ablation of heads i and j deviates from the sum of their individual effects. EAP edge scores (Syed et al., 2023) measure linearized gradient flow between heads. Comparing the magnitude-ranked pairwise Walsh coefficients to the corresponding EAP edge products gives Spearman $\rho = 0.58$ and Pearson $r = 0.73$. The two methods agree on which pairs interact most strongly — the top interacting pair by both measures is L8H6–L5H5, two name movers — but diverge on weaker pairs where nonlinear routing effects dominate. The moderate correlation supports the thesis that different discovery methods impose different selective pressures on the interaction structure they recover: Walsh coefficients capture the full nonlinear interaction, while EAP captures the linear component.

Scaling implications. For a circuit of 30 heads ($k = 465$, $N \approx 10^9$), recovery at $M/k = 1$ requires roughly 500 forward passes — a ten-minute GPU job rather than an impossible enumeration. At 50 heads ($k = 1275$), the cost is ~ 1300 forward passes. Sparse Walsh recovery transforms the interaction analysis from a small-circuit diagnostic into a scalable circuit discovery method, applicable wherever activation patching can identify a candidate head set.

3.2 The interaction spectrum varies systematically across tasks

The Walsh–Hadamard decomposition partitions each circuit’s total variance into contributions from each interaction order: order-1 (individual head effects), order-2 (pairwise interactions), and order-3+ (higher-order terms). We computed exhaustive 2^n coalition sweeps for 60 circuit–task–ablation combinations across six tasks in GPT-2 small: IOI, reverse token identification (RTI), greater-than, induction, subject-verb agreement

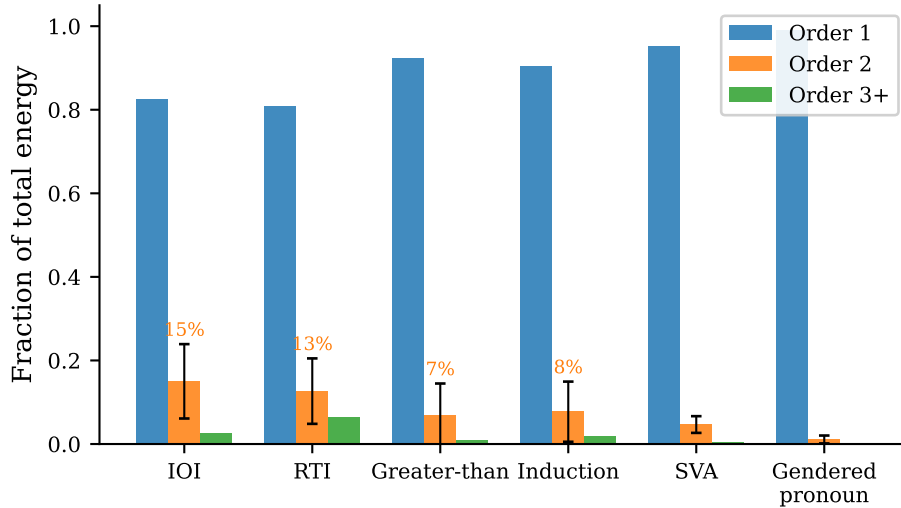


Figure 2: Walsh energy spectrum across six tasks in GPT-2 small, averaged over non-random circuits. IOI and RTI carry substantial order-2 energy (13–15%), indicating pairwise interactions that additive circuit summaries discard. Gendered pronoun resolution is nearly additive ($\sim 1\%$ order-2). Error bars show standard deviation across circuits within each task.

(SVA), and gendered pronoun resolution. Circuits range from 5 to 15 heads and include those discovered by activation patching, EAP, ACDC, Walsh-based selection, and random head sets.

Task	Circuits	Sweeps	Order-1	Order-2	Order-3+
IOI	7 (canonical, EAP, epistatic)	14	0.82	0.15	0.03
RTI	3 (incl. EAP)	6	0.81	0.13	0.06
Greater-than	5 (known, ACDC)	10	0.92	0.07	0.01
Induction	5 (known, ACDC)	10	0.90	0.08	0.02
SVA	5 (known, ACDC)	10	0.95	0.05	0.00
Gendered pronoun	5 (known, ACDC)	10	0.99	0.01	0.00
Random (all tasks)	6 random head sets	12	0.92	0.06	0.02

Energy fractions are averages over real (non-random) circuits within each task, rounded to two decimal places.

Three patterns emerge. First, the amount of pairwise interaction varies by task: IOI and RTI circuits carry 13–15% of their variance at order 2, while gendered pronoun circuits carry approximately 1%. The variation reflects the structure of the task: IOI requires coordinated attention routing across multiple head classes, whereas gendered pronoun resolution can be solved by a small set of independently operating heads.

Second, order-3+ energy rarely exceeds 5% for real circuits (5 of 48 non-random sweeps). The pairwise Walsh spectrum captures the dominant interaction structure; higher-order terms contribute marginally. This justifies focusing on order-2 coefficients: a sparse recovery of $k = n + \binom{n}{2}$ coefficients captures the interaction structure that matters.

Third, random head sets carry near-zero order-2 energy (average 6% across all tasks), confirming that the interaction structure of real circuits reflects functional coupling rather than an artifact of the measurement. The one exception is a random SVA circuit under zero ablation (35% order-2), where the ablation primitive creates spurious interactions by zeroing heads that the remaining heads depend on for LayerNorm scaling.

Figure 2 visualizes these patterns.

3.3 Weight geometry does not predict head-pair epistasis

All inter-head communication in a transformer passes through the residual stream, so heads that share writing and reading subspaces have a geometric reason to interact. If epistasis were a weight-level property, OV composition scores (Elhage et al., 2021) would predict pairwise Walsh coefficients, and coalition sweeps would be an expensive way to recover structure already present in the weight matrices. We tested the prediction directly.

Features and evaluation. Four predictors were computed from the LayerNorm-folded effective weights of GPT-2 small: OV subspace overlap (Frobenius cosine similarity of W_{OV} matrices), Q-composition and K-composition scores (Elhage et al., 2021), and layer distance. The target is the signed order-2 Walsh coefficient w_{ij} for each head pair, averaged over prompts. Because head pairs are dyadic — 105 pairs from 15 heads share endpoints and are therefore dependent — we cross-validated with a leave-both-heads-out procedure: for each test pair (i, j) , the model trains only on pairs (a, b) with $a, b \notin \{i, j\}$. Standard k -fold cross-validation leaks through shared heads and inflates R^2 . Uncertainty intervals use a delete-one-head jackknife.

Result. Across ten circuit-primitive combinations from three tasks (IOI, RTI, greater-than), all 30 cross-validated R^2 values are negative:

		zero	mean	resample
IOI (15 h, 512 p)	in-sample R^2	0.24	0.12	0.16
	CV R^2	−0.36	−0.00	−0.13
	split-half reliability	0.99	0.96	0.97
RTI median (15 h, 4 circuits)	in-sample R^2	0.05	0.06	0.10
	CV R^2	−0.13	−0.14	−0.20
	split-half reliability	0.94	0.97	0.94

The in-sample values reflect overfitting: 105 pairs from 15 heads give roughly 43 effective observations for 4 predictors. A layer-distance-only baseline is equally uninformative (CV $R^2 < 0$ under all primitives for both tasks). Five greater-than circuits (7 heads, 21 pairs each, 1000 prompts) confirm the pattern: every CV R^2 is negative (range −0.34 to −8.51), and their in-sample values (up to 0.63) sit within the null distribution at that sample size.¹

Interpretation. The failure is not a measurement problem. Split-half reliability of the Walsh coefficients exceeds 0.89 in 29 of 30 arms (the exception is one RTI random circuit under resample ablation, $r = 0.67$), so the target is precisely measured. Across ablation primitives on the same circuit, 32% of head pairs change the sign of their interaction between zero and mean ablation — precise measurement of quantities that systematically disagree (Housden et al., 2017). The cross-primitive disagreement is not an artifact of our decomposition. The Walsh–Hadamard transform already averages interaction coefficients over all 2^n head coalitions rather than anchoring to a single reference state (Poelwijk et al., 2016; 2019) — the standard approach for removing dependence on an arbitrary baseline. Quantitative genetics faces the same problem when measuring interactions between mutations: different loss-of-function assays yield different interaction estimates, and an active dispute over whether reference-free methods reduce or merely relabel higher-order structure remains unresolved (Park et al., 2024; Dupic et al., 2024; Buda et al., 2024). Our 32% sign-flip rate is comparable to rates observed in protein fitness landscapes, where ~35% of mutation pairs change interaction type depending on the measurement method (Baheti et al., 2024; Housden et al., 2017). Correlating each pair’s Walsh coefficient across primitives (Ferretti et al., 2016) gives $\gamma = 0.14$ between zero and mean or resample ablation, and $\gamma = 0.43$ between mean and resample. Mean and resample ablation, which both substitute realistic activations, agree moderately with each other; zero ablation, which removes the head’s contribution entirely, agrees with neither.

¹RTI arms use 302 or 512 prompts depending on the ablation primitive; results are flagged but qualitatively identical.

Weight geometry predicts interaction under no primitive. Head-pair epistasis is a property of the forward pass, shaped by data-dependent LayerNorm normalization and attention routing, that static weight matrices do not capture. The coalition sweep measures something the weights do not contain.

3.4 Path patching and Walsh interaction are nearly orthogonal

EAP edge scores correlate moderately with Walsh interaction coefficients ($\rho = 0.58$; Section 3.1). Path patching isolates a narrower quantity — the direct causal effect routed through a specific receiver — and might correlate more strongly with the nonlinear interaction or might diverge. We tested the correlation on the same 20-head IOI circuit.

Methods. For each of the 176 cross-layer head pairs (S, R) with $\text{layer}(S) < \text{layer}(R)$, we computed the direct path-patch effect: corrupt the sender’s `hook_z` with its mean across 200 IOI prompts, freeze all intermediate attention heads and MLPs to their clean values, and let the receiver and all downstream components compute naturally. The change in mean logit difference relative to the clean run is the direct effect of S on the output routed through R . Fourteen same-layer pairs serve as a sanity check (direct effect is exactly zero by architecture). The experiment was pre-registered before any path patching scores were computed.

Result. Walsh interaction coefficients and path-patch direct effects are nearly orthogonal:

	$ w_{ij} $ versus $ \text{PP} $	signed
Spearman ρ	0.21	—
Pearson r	0.13	0.07

The top Walsh pair (L5H5–L8H6, $w = 0.200$) ranks 24th by direct path-patch effect. The top path-patch edge (L10H7→L11H1, $\text{PP} = -1.96$) has a Walsh coefficient of -0.004 , consistent with zero interaction. The two rankings share none of their top-10 entries.

Mediation fraction. Among the 46 pairs in the top quartile of $|w_{ij}|$, 7 (15.9%) fall in the bottom quartile of $|\text{PP}|$ — pairs whose Walsh interaction is strong but whose direct causal edge is weak. These interactions are mediated: the functional coupling between the two heads is routed through intermediate components rather than through the direct residual-stream path.

Interpretation. Walsh coefficients measure total interaction: how much the joint ablation of two heads deviates from the sum of their individual ablations, averaged over all 2^{n-2} backgrounds. Path patching measures direct information flow: how much of the sender’s signal reaches the output specifically through the receiver, with all other paths frozen. The two quantities can diverge whenever a pair interacts primarily through a shared intermediary — a head that reads from one and writes to the other, or an MLP that transforms the combined signal.

Figure 3 shows the scatter: pairs cluster along both axes independently, with no diagonal structure. The top Walsh pair (L5H5–L8H6) and the top path-patch edge (L10H7→L11H1) occupy opposite corners.

The near-zero correlation confirms that interaction structure and information flow are distinct properties of a circuit. A discovery method that ranks edges by direct causal effect (path patching, ACDC) imposes a different selective pressure from one that ranks by total interaction (Walsh). Neither subsumes the other: path patching finds the wiring diagram, and Walsh finds the functional couplings. The choice of method determines which structure the circuit retains.

3.5 Walsh-ranked circuits are faithful

The preceding sections compare discovery methods by how they weight the interaction spectrum. A complementary question is whether Walsh-based rankings produce faithful circuits — subnetworks that

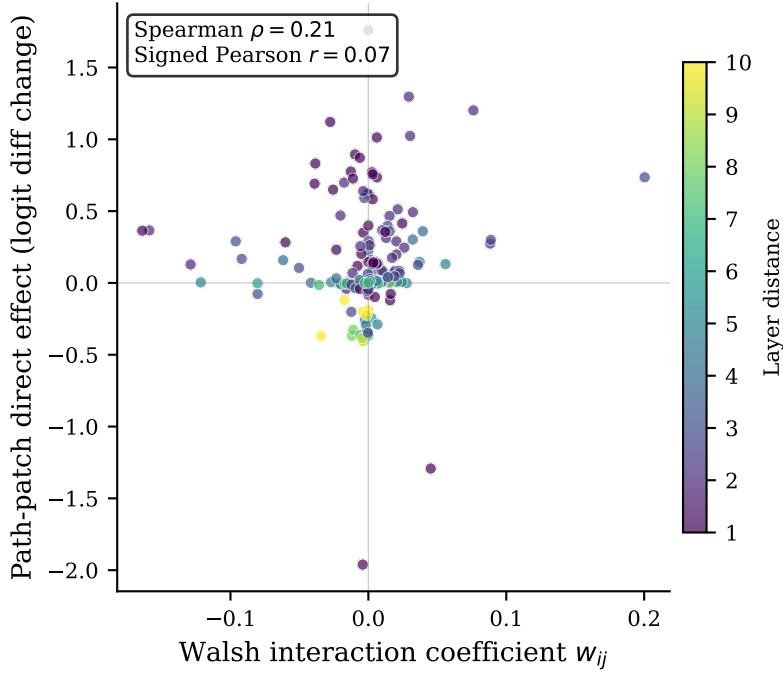


Figure 3: Walsh interaction coefficient versus path-patch direct effect for 176 cross-layer head pairs in the 20-head IOI circuit. Color indicates layer distance between sender and receiver. The two quantities are nearly uncorrelated (signed Pearson $r = 0.07$), confirming that functional coupling and direct information flow measure different aspects of circuit structure.

reproduce the full model’s behavior when the rest is ablated. We evaluated faithfulness using the MIB protocol (Mueller et al., 2025) on all 144 attention heads in GPT-2 small with IOI, measuring logit difference as the target metric and mean ablation of `hook_z` for heads outside the circuit.

Methods. Three head-ranking methods and a random baseline were compared at 13 circuit sizes from $k = 1$ to $k = 144$:

- **Walsh:** node importance for head h defined as $|w_{\{h\}}| + \sum_j |w_{\{h,j\}}|$, the order-1 magnitude plus the total pairwise interaction involving that head. Coefficients recovered by LASSO from $M = 12,500$ random coalitions (split-half $r = 0.994$, 1,002 nonzero coefficients out of 10,440).
- **Activation patching:** rank by $|\Delta\text{LD}|$ from single-head mean ablation.
- **Walsh + activation patching:** rank-sum combination of Walsh and activation patching rankings.
- **Random:** uniform random ranking, averaged over 5 seeds.

Faithfulness is computed with $m(\emptyset)$ normalization (Equation 3), where $m(N) = 4.378$ and $m(\emptyset) = 0.905$.

Result.	Method	CPR $\uparrow$	CMD $\downarrow$
	Walsh	0.996	0.100
	Activation patching	0.937	0.095
	Walsh + activation patching	0.956	0.104
	Random	0.608 ± 0.194	0.402 ± 0.180

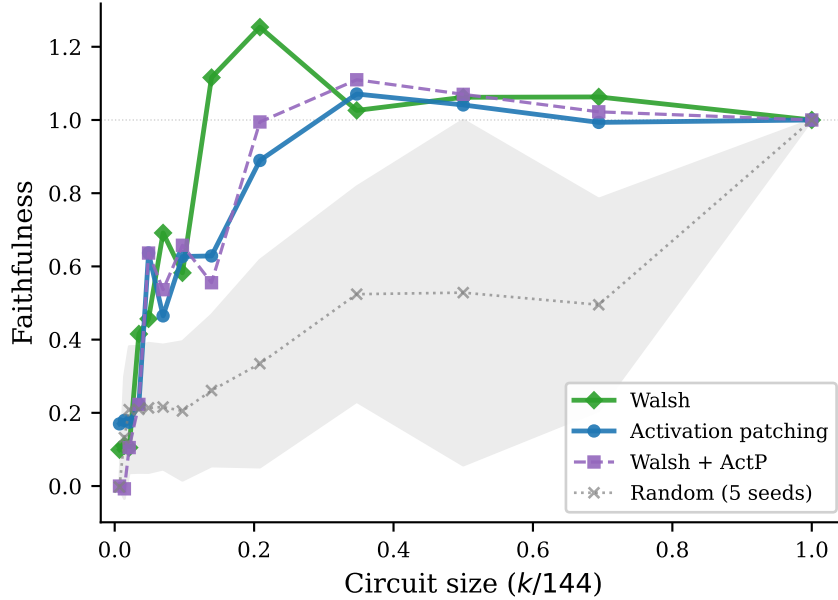


Figure 4: Faithfulness as a function of circuit size ($k/144$) for Walsh, activation patching, their rank-sum combination, and random (mean $\pm$ standard deviation over 5 seeds). Walsh (green diamonds) leads at small k and reaches $f > 1$ at 20 heads; activation patching (blue circles) converges more slowly. The gray band shows the random baseline. All structured methods converge by $k \approx 50$ heads.

Walsh-ranked circuits achieve the highest CPR (0.996), exceeding activation patching (0.937) by 0.059. Activation patching has the lowest CMD (0.095 versus 0.100 for Walsh): its circuits track the ideal $f = 1$ line more tightly, while Walsh circuits overshoot it at intermediate sizes. The rank-sum combination does not improve over either component (CPR = 0.956, CMD = 0.104), consistent with the two rankings identifying the same core heads through different routes — 16 of the top 20 Walsh-ranked heads also appear in the top 20 by activation patching.

This pattern matches the pair-level result from Section 3.4: Walsh and activation patching weight head pairs differently (interaction coefficients versus marginal effects), but that difference washes out when scores are aggregated to single-head importance. The node-level rankings are near-identical; combining them adds noise rather than complementary information.

Faithfulness curves. Walsh reaches $f > 1.0$ at $k = 20$ heads (14% of the model), where activation patching is still at $f = 0.63$. The gap is largest at small circuit sizes: at $k = 10$, Walsh achieves $f = 0.69$ while activation patching achieves $f = 0.46$. Both methods show non-monotonic faithfulness curves — adding certain heads decreases faithfulness before recovering at larger k — reflecting the epistatic interactions that this paper measures. All methods converge by $k \approx 50$ heads. Figure 4 shows the full curves.

4 What the interaction structure contains

The preceding sections rank discovery methods by the interaction structure they retain. That ranking earns its place only if the retained structure carries mechanism, so this section asks what is inside the order-2 spectrum of a circuit whose mechanism is already documented. Reading the canonical indirect object identification circuit of Wang et al. (2023) against its own taxonomy recovers a dependency between two of its head classes that the published account records as unexplained.

4.1 Double ablation decides whether two components are one unit

Genetics has decided whether two mutations belong to one unit or two since the 1950s, using a test that requires no knowledge of what the unit is made of. Two recessive mutations are placed in *trans*, and the phenotype of the double heterozygote decides one locus or two (Benzer, 1955). The reading is a sign convention on the interaction between the two perturbations:

interaction	reading	verdict
sub-additive (masking)	one pathway; the second hit has nothing left to break	same unit
additive	independent pathways	different units
super-additive (synergy)	parallel pathways, each covering for the other	different units

An exhaustive coalition sweep supplies exactly this data for every head pair at once. For a circuit of n heads, the 2^n coalition values determine every order-2 Walsh coefficient w_{ij} exactly, with no regularization and no held-out set. Taking $M_{ij} = -w_{ij}$ under the convention above yields a signed masking matrix over all $\binom{n}{2}$ pairs. Magnitude alone is uninterpretable here, since masking and synergy are both large $|w_{ij}|$; the sign carries the verdict.

Interpretability assigns head roles from what a head appears to do. Applying the complementation test to those roles asks whether the names track a functional criterion.

4.2 The taxonomy separates induction and name-mover heads, and merges two others

Hierarchical clustering on M for the 15-head IOI circuit, with cluster count selected by silhouette, recovers induction heads (L5H5, L6H9) and name-mover heads (L9H9, L10H0) as exact units. Adjusted Rand Index against the published labels is +0.571 as a point estimate, with a bootstrap-over-prompts 95% interval of $[+0.125, +0.571]$. The taxonomy therefore tracks interaction structure, by an amount the resampling leaves uncertain.

One boundary fails, and it fails stably. S-inhibition heads and negative name movers occupy a single cluster in **1000 of 1000** bootstrap resamples, while the global partition wobbles across the same draws. The merge holds in every robustness condition we ran: mean ablation as well as zero, circuits from EAP-IG and from order-1 Walsh discovery as well as the order-2 circuit, and complete and Ward linkage as well as average. Under a larger set of role labels the bootstrap merge rate falls to 81–86% on the zero-ablation conditions and remains at 100% on the others, so the pair separates occasionally once the clustering has more classes available to it.

Depth does not explain this. Roles in IOI are layer-stratified, so a clustering that recovered depth would score well against roles by accident. The derived partition matches role at ARI 0.571, raw layer at 0.053, and layer band at 0.217, while role itself matches layer band at 0.526. The clustering tracks role an order of magnitude better than depth.

4.3 The merge is asymmetric, and the asymmetry is the result

Figure 2 of Wang et al. (2023) places name movers, negative name movers, and backup name movers inside one group, with S-inhibition writing the queries of all three. On that topology, removing S-inhibition disables what the group does, and a merge between S-inhibition and any member follows.

The data merge one member and separate the other:

class pair	merged in
S-inhibition $\sim$ negative name mover	4 / 4 conditions
S-inhibition $\sim$ name mover	0 / 2 conditions
name mover $\sim$ negative name mover	0 / 2 conditions

Wang et al. (2023) state the only difference between the two classes: the negative heads “share all the same properties as Name Mover Heads except they (1) write in the opposite direction of names they attend to and (2) have a large negative copy score.” Same query source, same key and value input, same output edge, opposite OV sign. A topological account of the merge predicts that S-inhibition masks both classes, and the circuit diagram has no element that would distinguish them.

4.4 The negative heads invert rather than diminish

The OV sign difference predicts the asymmetry. Negative name movers write the negation of whatever they attend to. With S-inhibition intact they attend to the indirect object and write $-IO$, which lowers the logit difference. Removing S-inhibition redirects their attention to the subject, so they write $-S$, which *raises* the logit difference $IO - S$. Their contribution inverts. Name movers copy positively, so losing S-inhibition degrades them toward zero without changing their sign.

The exhaustive lattice tests this directly. Restricting to coalitions in which all S-inhibition heads are retained, and separately to those in which all are removed, the marginal contribution $\mathbb{E}_S[v(S \cup \{i\}) - v(S)]$ is exact in each condition:

class	head	S-inh. retained	S-inh. removed	sign flips
negative name mover	L10H7	−0.130	+0.137	1000 / 1000
negative name mover	L11H10	−0.057	+0.180	1000 / 1000
name mover	L9H9	+0.129	−0.012	878 / 1000
name mover	L10H0	+0.149	+0.009	318 / 1000

Figure 5 visualizes the pattern. Both negative heads invert in every bootstrap draw, moving from clearly harmful to clearly helpful. The name movers decay toward zero: L10H0 keeps its sign, and L9H9 crosses zero at a magnitude of 0.012, which is indistinguishable from no contribution at all. Sign reversal and neutralization are distinct outcomes. They produce the two interaction signs the clustering found: S-inhibition with negative name movers gives -0.267 (masking, one unit), while S-inhibition with name movers gives $+0.141$ (synergy, two units).

Wang et al. (2023) observed a fragment of this effect under a different conditioning set. After knocking out name movers, they report that “the Negative Name Mover Heads have a less negative effect on logit difference, and 10.7 even has a positive effect on the logit difference after the knockout,” and conclude that “[b]oth the reason and the mechanism of this compensation effect are still unclear.” Path patching measures one edge against a fixed background, which makes a dependency of this kind visible only where the background happens to expose it. The interaction decomposition holds the whole lattice, so the dependency appears for both heads, with a magnitude and a stability attached.

4.5 What an additive circuit discards

This dependency lives entirely in order-2. A circuit summarized by its order-1 coefficients retains the fact that negative name movers lower the logit difference on average, and loses the fact that the sign of their contribution is set by an upstream class. The EAP-IG circuit carries 90.4% of its energy at order 1 against the canonical circuit’s 78.6% (Table 1), and that gap is the room in which effects like the sign inversion sit.

The consequence for practice is narrow and concrete. A head’s measured contribution is a property of the head together with the background it was measured against, and reporting it as a property of the head alone is safe only when the order-2 terms coupling it to the rest of the circuit are small. Those terms are measurable, so the assumption is checkable rather than assumed.

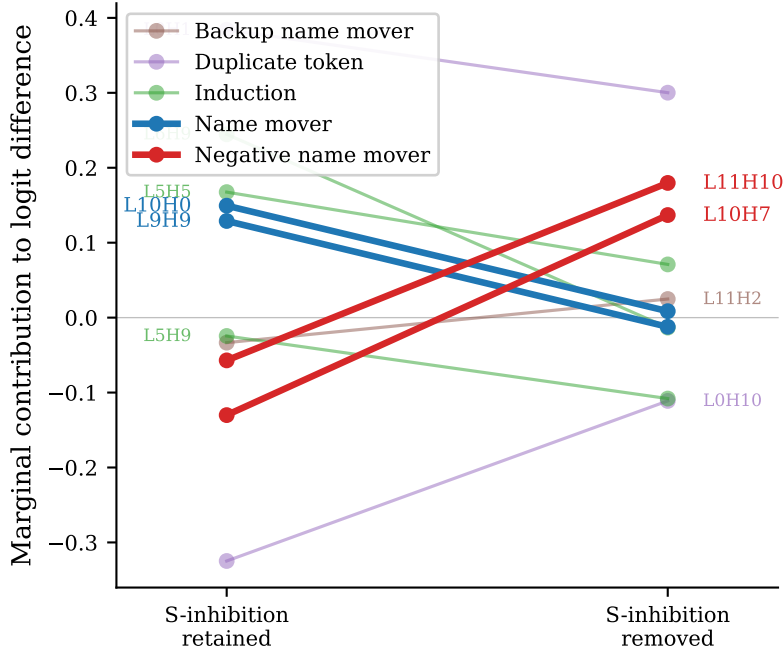


Figure 5: Marginal contribution of each head to logit difference, conditioned on whether S-inhibition heads are retained (left) or removed (right). The two negative name movers (red) cross zero, inverting from harmful to helpful. Name movers (blue) decay toward zero without inverting. Other head classes show moderate attenuation. Bootstrap sign-flip rates: both negative name movers flip in 1000/1000 draws.

Circuit	Order 1 (%)	Order 2 (%)	Order 3+ (%)
EAP-ranked	90.4	9.0	0.6
Canonical (Wang)	78.6	18.4	3.0
Walsh-ranked	80.9	16.8	2.3
Epistasis-ranked	78.4	18.6	3.0
Random	97.9	2.1	0.0

Table 1: Fraction of nonconstant Walsh energy at each interaction order for five 15-head IOI circuits under mean ablation. The EAP-ranked circuit concentrates energy at order 1; circuits selected for functional relevance (canonical, Walsh, epistasis) retain 17–19% at order 2.

5 Discussion

5.1 The selective-pressure analogy

Circuit discovery methods are evaluation instruments, and every instrument selects for what it measures. Activation patching selects for heads whose individual removal changes the target metric. EAP selects for edges along which gradients flow. Path patching selects for direct causal influence through the residual stream. Each method returns a “circuit” that is faithful to the full model, but the circuits differ in which interaction structure they preserve.

The Walsh–Hadamard decomposition provides a fixed reference frame for quantifying these differences. The interaction spectrum of a circuit on a task is a property of the model, independent of how the circuit was discovered. Two methods that return the same circuit preserve the same spectrum; two methods whose head rankings correlate at $\rho = 0.58$ (EAP) versus $\rho = 0.07$ (path patching) preserve detectably different structure.

The “right” discovery method depends on what the analyst wants to learn: marginal importance, direct wiring, or functional coupling.

5.2 Implications for circuit interpretation

The complementation test in Section 4 shows that interaction structure carries mechanism. The sign inversion of negative name movers is invisible to any method that reports only marginal importance: the head’s average contribution is negative, and its contribution conditional on the state of upstream S-inhibition heads flips sign entirely. Reporting the head’s marginal score as its property elides the dependency that determines whether the head helps or hurts.

This is a general warning, not specific to IOI. A head whose contribution depends on the state of another head is the rule rather than the exception whenever order-2 energy exceeds a few percent of the total spectrum. The fraction of energy at each order is measurable from the Walsh spectrum, so the severity of the problem is quantifiable for any circuit.

5.3 Limitations

One model, one task. All results use GPT-2 small on IOI. The Walsh–Hadamard transform applies to any model and task where components can be ablated and a scalar metric computed, but the empirical findings — the correlation between Walsh and EAP, the orthogonality of Walsh and path patching, the complementation-test result — may not generalize. The compressed sensing protocol makes it feasible to run the same analysis on larger models and different tasks.

Heads only. The current decomposition treats attention heads as the atomic components. MLPs, individual neurons, and attention positions are excluded. Extending to finer granularity increases n and the number of target coefficients, but the compressed sensing approach scales as $O(k \log k \log N)$, not $O(2^n)$, so moderate increases are tractable.

Ablation dependence. The Walsh spectrum depends on the ablation primitive (Section 3.3). Mean and resample ablation agree moderately ($\gamma = 0.43$); zero ablation agrees with neither ($\gamma = 0.14$). The interaction structure is not a unique property of the circuit — it is a property of the circuit together with the measurement method. This parallels the situation in quantitative genetics, where different loss-of-function assays yield different interaction estimates (Housden et al., 2017; Baheti et al., 2024).

Faithfulness metric choice. MIB faithfulness (Section 3.5) measures how well a circuit reproduces the full model’s logit difference under mean ablation. Chughtai et al. (2024) show that faithfulness metrics are not robust to the choice of ablation primitive or metric. Our results should be interpreted as specific to mean ablation on logit difference, not as general claims about circuit quality.

5.4 Future directions

The sparse Walsh recovery protocol converts interaction analysis from a small-circuit diagnostic into a practical method applicable wherever activation patching can identify a candidate head set. Running the analysis on multiple models (GPT-2 medium, Pythia, Llama) and tasks (greater-than, gendered pronoun resolution) would test whether the selective-pressure pattern generalizes. Extending the decomposition to MLP neurons alongside attention heads would capture cross-component interactions that the head-only analysis misses. Combining Walsh interaction rankings with EAP edge scores — rather than with path patching, which measures an orthogonal quantity — may produce circuits that are both faithful and interaction-preserving.

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